

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1552

Bedford 138 kV
January 25, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1552. The Interconnection Customer proposes a 94.8 MW battery energy storage facility interconnecting at the Bedford 138 kV substation in the Kentucky area. The request was received on August 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 503 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,656,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Bluestem-Bodo 230 KV: 136.4% loading (498 MVA) following the outage of the Osawatomie-Tmarisk 230 KV L; DFAX = 5.8%.

Hot Springs-Southside Wastewater Treatment Plant 100 KV line: 121.4% loading (216 MVA) following the outage of the Bluestem-Bodo 230 KV Line; DFAX = 14.4%.

For the loss of Carraizo 1201-Pine Bluff Watson Chapel 345 KV, the Eldorado 345/138kV Auto loads to 117.7% of its 402 MVA rating. Project DFAX: 15.8%.

Brincken'S Corner-Autaugaville 100 KV: 117.6% loading (167 MVA) following the outage of the Bluestem-Bodo 230 KV Line; DFAX = 13.0%.

The Lg&E; Canal-Hopson 100 KV #1 is overloaded to 114.1% of its 155 MVA emergency rating for the loss of the Lg&E; Canal-Hopson 100 KV. The project contributes 5.7% DFAX (5.4 MW) to this overload.

For the loss of Lg&E; Canal-Hopson Substation 100 KV #1, the Lg&E; Canal-Hopson 100 KV loads to 114.1% of its 155 MVA rating. Project DFAX: 5.7%.

Southside Wastewater Treatment Plant-Osawatomie 100 KV: 112.1% loading (249 MVA) following the outage of the Bluestem-Bodo 230 KV L; DFAX = 10.8%.

The Maracaibo Substation-University Substation 138 KV is overloaded to 110.5% of its 255 MVA emergency rating for the loss of the Pine Bluff Watson Chapel-Carraizo 1201 345 KV. The project contributes 31.9% DFAX (30.2 MW) to this overload.

Hot Springs-Irick 100 KV: 107.2% loading (207 MVA) following the outage of the Bluestem-Bodo 230 KV line; DFAX = 14.4%.

For the loss of Bodo-Bluestem Substation 230 KV Line, the Finley Combustion-Prager 138 KV line loads to 105.7% of its 264 MVA rating. Project DFAX: 5.7%.

Carraizo 1201-Pine Bluff Watson Chapel 345 KV: 104.0% loading (1,227 MVA) following the outage of the Lee County Solid Waste Energy Substation-Cooper 345 KV; DFAX = 13.6%.

The Picnic Point-Oakland Ave 230 KV is overloaded to 101.8% of its 628 MVA emergency rating for the loss of the Mexico Intertie-Newberry Vlg 230 KV L. The project contributes 8.2% DFAX (7.8 MW) to this overload.

Firstenergy Albright-Bedford 138 KV: 101.6% loading (201 MVA) following the outage of the Pine Bluff Watson Chapel-Carraizo 1201 Substation 345 KV; DFAX = 57.8%.

For the loss of Tamarisk-Osawatomie 230 KV line, the Bluestm 230 kv/1 kV TX loads to 101.1% of its 634 MVA rating. Project DFAX: 5.8%.

The Schertz-Sierra Pine 138 KV is overloaded to 100.2% of its 209 MVA emergency rating for the loss of the Sierra Pine-Carntown 138 KV. The project contributes 8.9% DFAX (8.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bluestem Substation-Bodo 230 KV	Rebuild line with higher-capacity conductor	\$14,876,000	5% (\$704,000)
Hot Springs-Southside Wastewater Treatment Plant 100 KV	Rebuild line with higher-capacity conductor	\$841,000	3% (\$25,000)
Eldorado Substation 345/138kV Auto	Replace transformer with higher-rated unit	\$18,453,000	4% (\$763,000)
Autaugaville-Brincken'S Corner 100 KV	Rebuild line with higher-capacity conductor	\$2,262,000	3% (\$66,000)
Lg&E; Canal-Hopson 100 KV #1	Rebuild line with higher-capacity conductor	\$462,000	14% (\$64,000)
Lg&E; Canal-Hopson 100 KV	Rebuild line with higher-capacity conductor	\$387,000	14% (\$54,000)
Southside Wastewater Treatment Plant-Osawatomie 100 KV	Rebuild line with higher-capacity conductor	\$1,815,000	3% (\$59,000)
Maracaibo Substation-Universiity Substation 138 KV	Rebuild line with higher-capacity conductor	\$3,931,000	16% (\$622,000)
Irick-Hot Springs 100 KV Line	Reconductor line	\$813,000	3% (\$24,000)
Finley Combustion-Prager 138 KV line	Reconductor line	\$1,140,000	6% (\$68,000)
Carraizo 1201-Pine Bluff Watson Chapel 345 KV	Reconductor line	\$12,084,000	2% (\$250,000)
Picnic Point-Oakland Ave 230 KV	Reconductor line	\$7,286,000	3% (\$223,000)
Firstenergy Albright-Bedford 138 KV	Reconductor line	\$12,136,000	34% (\$4,170,000)
Bluestem 230-1 kV Auto	Replace transformer with higher-rated unit	\$6,991,000	5% (\$331,000)
Schertz-Sierra Pine 138 KV	Reconductor line	\$3,776,000	6% (\$233,000)
	Total allocated to project		\$7,656,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.